

LANGOE, Denmark

WMO: 061380

Lat: 54.821N

Lon: 10.994E

Elev: 13

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.0	24.9	16.6	12.7	23.9	19.8	14.9	26.4	59.1	41.8	51.7	41.4	14.8	120	0.771

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	78.1	66.8	74.9	66.0	72.2	65.0	69.1	75.1	67.3	72.5	65.8	70.8	12.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.9	98.9	72.1	65.2	93.3	70.3	63.7	88.4	68.8	33.2	75.1	31.8	72.6	30.6	71.0	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 41.6	34.0	29.9	DB	20.2	83.2	3.7	3.9	17.6	86.1	15.4	88.3	13.3	90.5	10.7	93.4	(4)
(5)			WB	18.9	71.7	3.5	2.8	16.4	73.8	14.4	75.4	12.4	77.0	9.9	79.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	35.4	34.9	39.0	46.5	54.3	60.2	65.2	65.1	59.5	51.5	43.7	38.3		
	DBStd	11.93	5.55	5.49	4.98	5.13	4.94	4.26	4.85	4.07	3.96	4.60	4.91	5.71		
	HDD50	1954	452	422	341	131	14	0	0	0	0	38	194	362		
	HDD65	5768	917	842	805	557	333	153	59	48	171	417	639	827		
	CDD50	1795	0	0	0	25	148	307	470	469	285	86	5	0		
	CDD65	134	0	0	0	0	2	10	64	52	6	0	0	0		
	CDH74	364	0	0	0	0	5	26	197	129	7	0	0	0		
	CDH80	44	0	0	0	0	0	3	25	16	0	0	0	0		
	WSAvg	16.0	18.3	17.1	16.3	15.0	14.8	14.8	13.5	13.9	15.1	16.5	17.7	18.9		
	PrecAvg	22.7	1.9	1.5	1.5	1.4	1.5	2.1	2.1	2.5	2.2	2.1	2.0	1.9		
(15) Precipitation	PrecMax	28.1	4.1	3.3	3.5	3.3	2.8	4.4	4.3	5.3	5.2	4.2	5.0	3.8		
	PrecMin	16.3	0.0	0.1	0.3	0.6	0.5	0.0	0.5	0.6	0.4	0.7	0.4	0.9		
	PrecStd	3.7	1.1	0.7	0.8	0.6	0.6	1.2	1.1	1.4	1.6	0.9	1.1	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.9	46.7	53.7	66.6	73.3	77.9	83.5	81.9	74.7	63.7	56.2	49.8		
		MCWB	46.2	44.0	48.3	56.1	62.0	67.6	68.8	66.1	63.7	59.7	54.4	48.4		
	2%	DB	45.5	44.6	49.0	60.5	67.9	72.7	79.1	77.6	70.3	60.7	53.7	48.0		
		MCWB	43.7	42.8	45.8	52.8	59.7	64.9	67.4	65.8	62.7	57.7	52.4	46.6		
	5%	DB	44.0	43.0	46.4	56.7	64.4	69.5	75.8	74.0	67.3	59.0	51.7	46.7		
		MCWB	42.5	41.6	43.8	51.0	57.5	63.1	66.9	65.6	61.8	56.6	49.9	45.6		
	10%	DB	42.5	41.8	44.8	53.9	61.5	66.8	73.0	71.2	65.0	57.5	50.1	45.2		
		MCWB	41.3	40.6	42.7	49.5	56.1	61.3	65.7	64.8	60.7	55.4	48.5	44.1		
	0.4%	WB	46.6	44.6	48.7	57.5	64.5	69.4	71.9	70.6	66.5	60.7	54.9	49.0		
		MCDB	47.6	45.9	52.0	63.4	71.1	76.3	77.8	76.1	71.4	62.8	55.7	49.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.1	43.1	46.2	54.1	61.0	66.1	70.0	68.0	64.1	58.4	52.7	47.0		
		MCDB	45.0	44.2	48.7	59.0	65.7	70.9	76.0	73.0	67.8	60.0	53.6	47.6		
	5%	WB	42.8	41.9	44.4	51.9	58.7	64.1	68.2	66.6	62.6	57.0	50.3	45.7		
		MCDB	43.8	42.9	45.9	55.5	62.7	68.3	73.7	71.6	66.4	58.7	51.4	46.4		
	10%	WB	41.4	40.7	43.1	50.0	56.8	62.1	66.4	65.4	61.2	55.7	48.6	44.3		
		MCDB	42.4	41.7	44.5	53.2	60.6	66.1	71.8	70.2	64.5	57.2	49.9	45.2		
	5% DB	MDBR	4.3	4.4	5.9	8.5	9.0	9.0	9.5	8.4	6.9	5.4	4.4	4.3		
		MCDBR	4.7	5.1	8.8	13.7	13.7	13.9	14.3	13.4	11.0	6.8	5.0	4.4		
		MCWBR	4.6	4.3	6.4	8.0	8.4	8.6	6.6	5.7	6.0	5.2	5.0	4.6		
		MCWBR	4.6	5.0	8.2	12.5	12.0	12.7	12.6	10.8	9.7	6.4	4.8	4.3		
(35) Mean Daily Temperature Range	5% WB	MCWBR	4.7	4.4	6.4	7.9	8.1	8.8	6.7	6.0	6.1	5.3	4.9	4.4		
	taub		0.320	0.341	0.369	0.404	0.396	0.391	0.411	0.411	0.383	0.360	0.339	0.310		
		taud	2.295	2.314	2.272	2.225	2.337	2.379	2.341	2.350	2.407	2.417	2.351	2.281		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		193	233	252	259	269	272	263	255	247	224	184	168		
		Edh at Noon	18	25	32	39	37	36	36	34	29	23	17	15		
	RadAvg		151	346	774	1315	1679	1788	1665	1337	961	494	197	108		
		RadStd	23	57	73	158	177	122	186	92	94	63	27	8		

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (63 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	-288	N/A	N/A	(47)

Nomenclature: See separate page